Demographics

Overview

Here is my initial exploration of participant demographics.

Here are my research questions:
- How many respondents took the survey, and from how many institutions?
- Which regions are those institutions located in?
- What roles do our respondents represent?
- What are our respondents' professional backgrounds?

Import packages, utilities, and config variables

```
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(docstring)
```

```
Attaching package: 'docstring'

The following object is masked from 'package:utils':

    ?
```

```
library(readr)
```

```
Warning: package 'readr' was built under R version 4.4.3
```

```
source("utils.R")

# Read config.yml
cfg <- config::get()
```

Load data

```
mc_complex <- read_tsv(file.path(cfg$data_path, "mc_complex.tsv"), show_col_types = FALSE)
mc_other <- read_tsv(file.path(cfg$data_path, "mc_other.tsv"), show_col_types = FALSE)
prof_bg <- read_tsv(file.path(cfg$data_path, "prof_bg.tsv"), show_col_types = FALSE)
sata_other <- read_tsv(file.path(cfg$data_path, "sata_other.tsv"), show_col_types = FALSE)
qual <- read_tsv(file.path(cfg$data_path, "qual.tsv"), show_col_types = FALSE)
qual_pii <- read_tsv(file.path(cfg$data_path, "qual_pii.tsv"), show_col_types = FALSE)
```

How many respondents took the survey?

```
nrow(mc_complex)
```

```
[1] 25
```

```
nrow(prof_bg)
```

```
[1] 25
```

The numbers above^ should match, and this represents the number of respondents.

How many institutions?

I'm commenting out the relevant code for privacy reasons, but I first manually inspected the responses.

```
# Keep commented for privacy!!
# qual_pii$institution
```

Looking at the responses, I see several "categories" of institution: Network, networked university campus, independent university campus, and university-affiliated research center. "Networked" refers specifically to an OSPO network, not any other kind of network. Grouping the institutions by category, rather than listing them, will help protect respondent anonymity.

I have manually cleaned up the responses (a lot of people listed their OSPO/department as well as their institution) and listed the responses, categories, and regions in data/manually_curated/orgs.tsv.

```
# Do not preview in notebook, for privacy!!
orgs <- read_tsv(
  file.path(cfg$curated_path, "orgs.tsv"),
  show_col_types = FALSE,
  col_names = c("org", "cat", "region")
)

orgs <- orgs |>
  count(across(all_of(c("org", "cat", "region"))), name = "cnt")

# This is the number of unique institutions
nrow(orgs)
```

```
[1] 22
```

```r
orgs <- orgs |>
  arrange(desc(cnt))

# Write to file (from utils.R)
write_df_to_file(orgs, "orgs_summary.tsv", path_key = "curated_path")

# Summary of org categories
table(orgs$cat)
```

```
        Independent University Campus                                    Network
                                   13                                          1
          Networked University Campus University-affiliated Research Center
                                    5                                          3
```

Which regions are those institutions located in?

```r
table(orgs$region)
```

```
Europe     US
     6     16
```

^The table above shows represented regions.

Examining what people said for “other”

```r
# Keep commented for privacy!!
# mc_other$region_other
```

What roles do our respondents represent?

```r
role_label <- "role_mc"
table(mc_complex[[role_label]])
```

```
   I am a staff member of an OSPO or equivalent function
                                                         5
I carry out OSPO-like activities as part of another role
                                                         3
        I lead or co-lead an OSPO or equivalent function
                                                        16
                                                     Other
                                                         1
```

Examining what people said for "other"

```
# Keep commented for privacy!!
# mc_other$role_other
```

What about job titles? Need to code these. Writing to a file for this purpose.

```
write_df_to_file(qual$job, "job_titles.txt", path_key = "coding_path")
```

What are our respondents' professional backgrounds?

```
# Returns a list containing the frequency table for each column
table_list <- lapply(prof_bg, table)

# Examined the output by eye to make sure there's no weirdness
# table_list

# Extract the "Yes" count for each professional background
# This was a SATA question so total != # of respondents
yes_counts <- sapply(table_list, function(x) x["Yes"])
yes_counts <- sort(yes_counts, decreasing = TRUE)
print(yes_counts)
```

```
  faculty.Yes       rse.Yes        it.Yes     admin.Yes  industry.Yes
            8             8             6             5             5
     tto.Yes     legal.Yes community.Yes   library.Yes
            4             3             2             1
```

Examining the “other” responses. Mostly it looks like people just elaborating on the responses above, though one person said they were previously a volunteer to OS projects, which is interesting.

```
# Keep commented for privacy!!
# sata_other$prof_bg_other
```